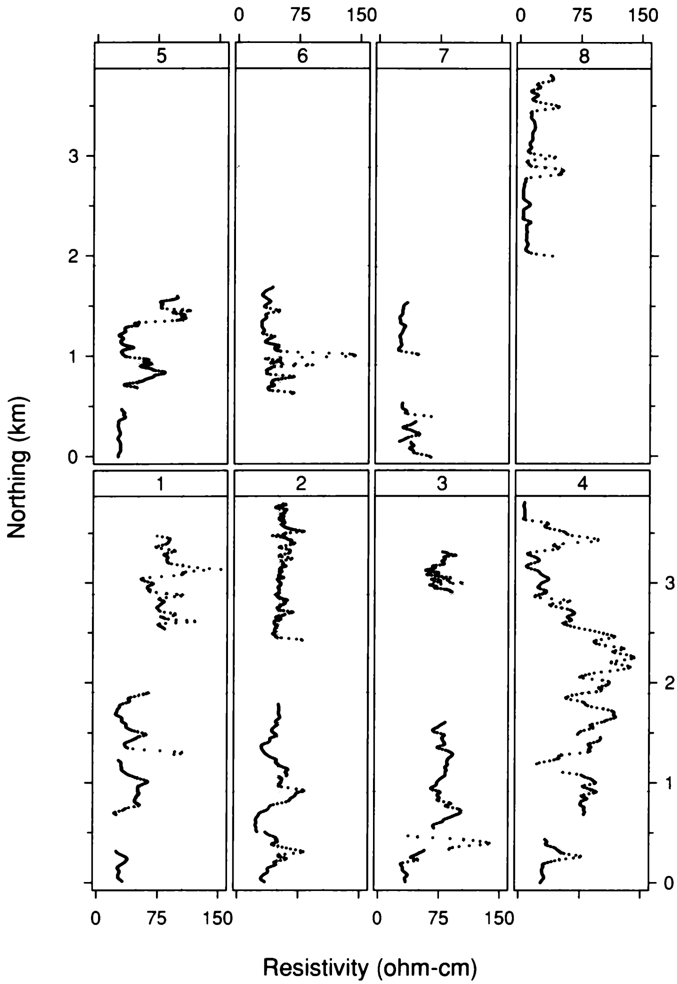
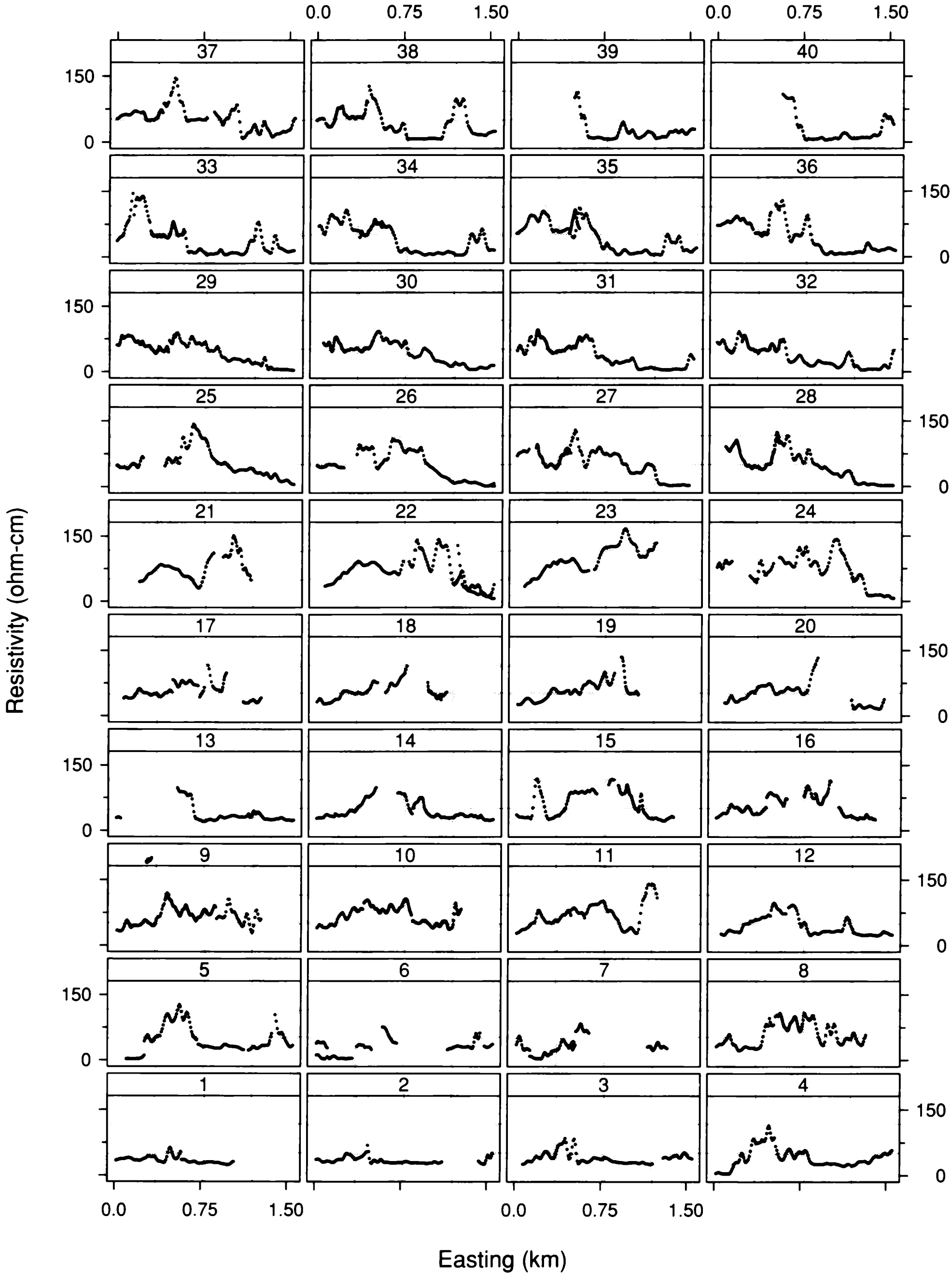


4.65 Resistivity is graphed against the northing coordinate for the eight tracks that run along the northing axis.



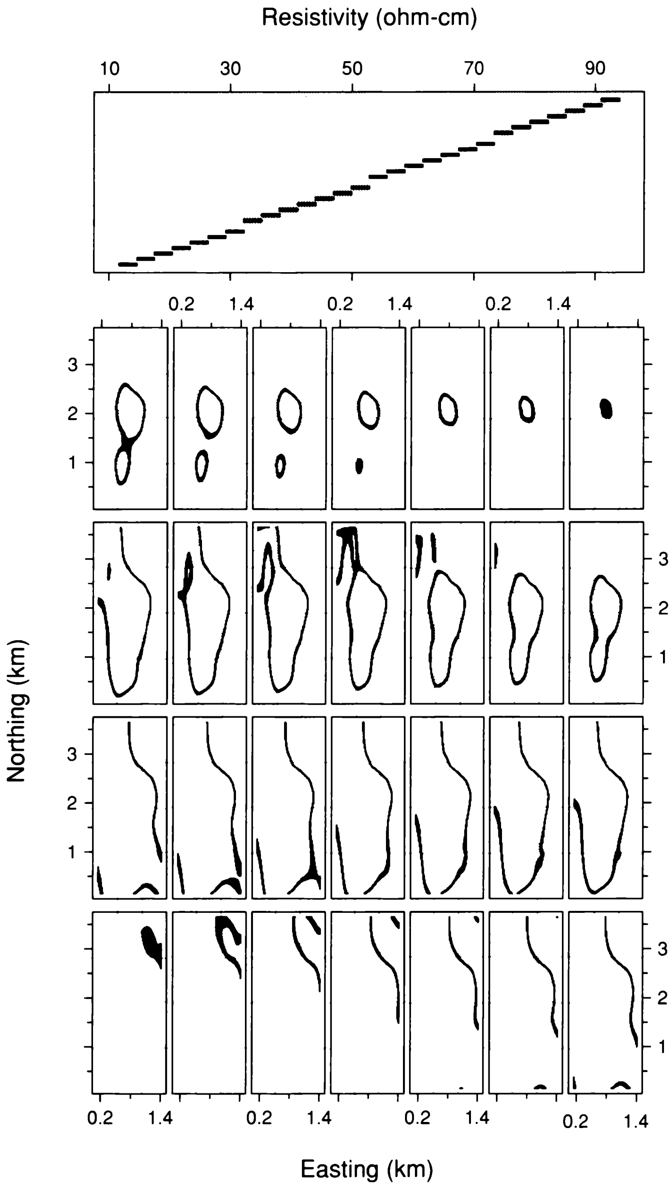
4.66 Resistivity is graphed against the easting coordinate for the 40 tracks that run along the easting axis.

Fitting

For the soil data, one important goal is to understand trends in the data to determine the broad areas where the overall level of salinity is high. We will do this with a loess trend fit with parameters $\alpha = 1/4$ and $\lambda = 2$. To preserve the natural metric of the spatial factors, the easting and northing coordinates will not be standardized by dividing by the 10% trimmed standard deviations. Also, bisquare will not be added because the visualization of the measurements in Figures 4.65 and 4.66 does not suggest outliers or other destructive behavior.

Figure 4.67 is a level plot of the loess fit evaluated on an 85×234 grid with 15m spacing ranging from 150m to 1410m along the easting axis and from 150m to 3645m along the northing axis. The grid is set slightly inside the rectangle formed from the extremes of the measurement locations. Such cropping is discussed in Chapter 5. The level plot has 28 equal intervals of resistivity that range from the minimum of the function values over the grid to the maximum; the only overlap of the intervals occurs at the endpoints.

Figure 4.67 shows that the lowest levels of resistivity occur near the upper right corner of the measurement region. From this depression, the level regions spread out to the extremes of the region, and then move inward to form two peaks, the highest one very near the center of the region.



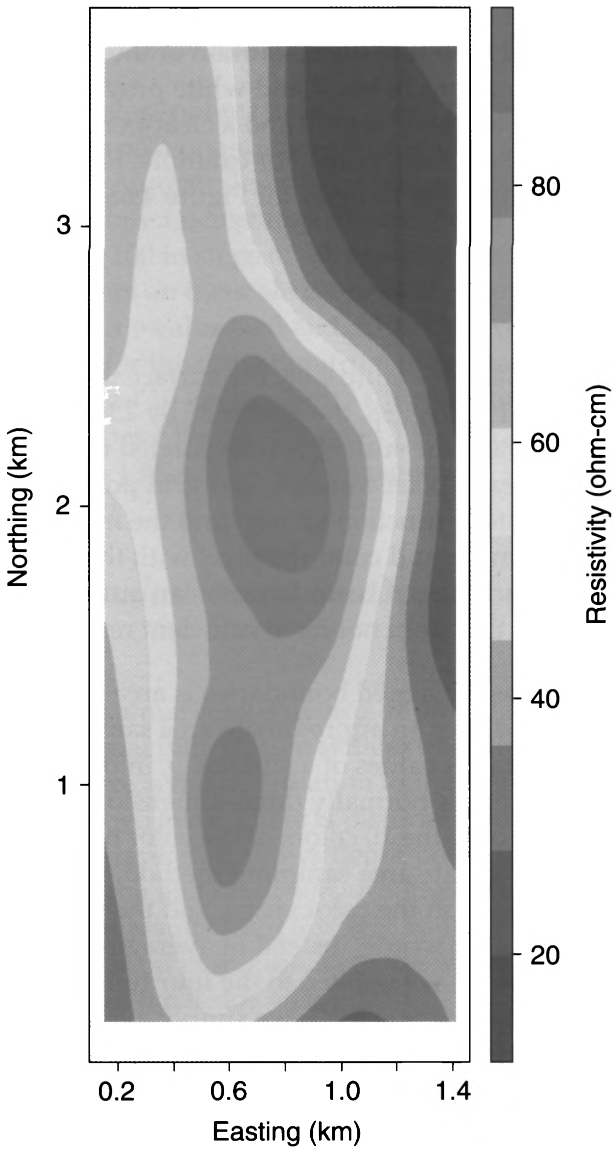
4.67 A level plot displays the loess fit to the soil data. The aspect ratio of the level-region panels has been chosen to make the number of data units per cm the same on both scales.

Superposed Level Regions in Color

The level plot in Figure 4.67 shows the level regions by juxtaposing them on different panels with the same scales. Using color, the level regions can be superposed. This is illustrated in Figure 4.68. In this case there are only 10 intervals, ranging from the minimum to the maximum function value. The wide use of level plots in color might suggest that the method is easy, but it is easy only if one is satisfied with a poor choice of colors such as a rainbow encoding. As Tufte puts it [75], "the mind's eye does not readily give an order to ROYGBIV."

There are two desiderata in choosing a color encoding of the quantitative values of a function. First, we typically want effortless perception of the order of the values. For example, effortless perception means we do not have to constantly refer to a key. Second, we want clearly perceived boundaries between adjacent levels. Achieving these two desiderata is difficult because they play against one another; it is easy to achieve one or the other, but hard to achieve both simultaneously.

The color encoding in Figure 4.68, which we will call THVL (two hues, varying lightness), represents a good compromise [54, 75]. First, there are two hues. Because the end product in this case is on a book page, the hues are cyan and magenta, which are two of the four colors standardly used in printing on paper — cyan, yellow, magenta, and black. The lightness of the colors decreases as the encoded values move away from a central value. From the middle to the extremes, the cyan ranges from 20% cyan to 100% cyan in steps of 20% cyan, and the magenta ranges from 20% magenta to 100% magenta in steps of 20% magenta. Using two hues and changing just their lightness provides the effortless perception of order. But there is a bound on the number of intervals that can be used if the desideratum of distinct boundaries is to be achieved. Because of the delicacy of color reproduction, only 10 have been used in Figure 4.68. At a computer screen it is possible to drive the number up to 15 or so, but using a significantly greater number typically results in a perceptual merging of some of the adjacent colors. Actually, more than 10 colors are used in Figure 4.68, but the additional ones occur just at boundaries of level regions and are only barely perceptible. This *anti-aliasing*, which is described later in a section for the record, gives the boundaries a smooth look. The smoothing was necessary because the display started out as an image on a computer screen; the low resolution of screens results in jagged boundaries unless anti-aliasing methods are used.



4.68 The fitted resistivity surface is displayed by a level plot with ten level regions superposed in color. An anti-aliasing method based on approximate area sampling gives the boundaries of the level regions a smooth appearance. The aspect ratio of the level-region panels has been chosen to make the number of data units per cm the same on both scales.

Superposing level regions in color rather than juxtaposing them in black and white has one benefit. We are better able to perceive the relative locations of the level regions, and this often conveys the overall gestalt of the surface more effectively. But the disadvantage is the limit to the number of colors and therefore the resolution of the display. A level plot with juxtaposed regions in black and white provides both a clear visual delineation of each level region and a clear perception of order even for a large number of intervals. For example, Figure 4.67 has 28 intervals, almost three times the number for Figure 4.68.

For the Record: Anti-Aliasing

In the anti-aliasing method of Figure 4.68, the surface is computed over a rectangular grid. In Figure 4.68, the grid is 85 by 234, the same grid used in Figure 4.67. Then, the surface is interpolated to a much denser grid, set up so that each pixel contains four grid points. For each pixel, we could average its four function values, find the interval among the 10 that contains the average, and color the pixel with the color for the interval. This would produce jagged boundaries on an output device, such as a computer screen, that does not have sufficient resolution.

One method for anti-aliasing jagged boundaries is area sampling [42]. The rendering in Figure 4.68 uses an approximate area sampling scheme. Let $c(y)$ be a function that takes values of a surface, y, into a color-description space, which is normally three-dimensional [42, 79]. Suppose that $c(y)$ takes on only the 10 colors of the color bar on the right in Figure 4.68. In the approximate area sampling method, the four colors of $c(y)$ for a pixel are averaged in the color space, and the pixel is filled with the average color. In other words, instead of averaging the four function values and applying c, we apply c to the four function values and average the colors. If the space is RBG, we simply average the intensities of the four reds, the four blues, and the four greens. Sometimes, though, the averages can produce colors that contrast with their neighbors, in which case we must alter the colors of the averages to maintain a smooth transition; an example will be given shortly. For most pixels, the four colors are the same, so the average is one of the 10 values of $c(y)$. But if a pixel is at a boundary between level regions, the colors are not the same, and the pixel receives, typically, a color that is not one of the 10. In Figure 4.68 the overall gestalt appears to have only the 10 colors of $c(y)$ because there is so little filling with other colors, but the small amount of additional coloring produces smooth boundaries by providing a smooth perceptual transition from one region to the next.

The $c(y)$ for the 10 colors of the bar in Figure 4.68 can be defined in a particularly simple way on a one-dimensional scale. Let $c(y)$ range from -100 to 100. A positive number describes a percentage of cyan and a negative number describes a percentage of magenta. The colors of the bar range from -100 to 100 in steps of 20, except there is no zero. The average of any four colors on the scale is one of the numbers -100 to 100 in steps of 5, except for the values ±5 and ±15. The averages work quite well in this case except for the transition from magenta to cyan; the values -10, 0 (white), and 10 are too light. The following alteration was used in Figure 4.68: -10 is 13% magenta and 7% cyan; 0 is 10% magenta and 10% cyan; and 10 is 13% cyan and 7% magenta.

4.13 Direct Manipulation and Shading for 3-D Plots of Surfaces

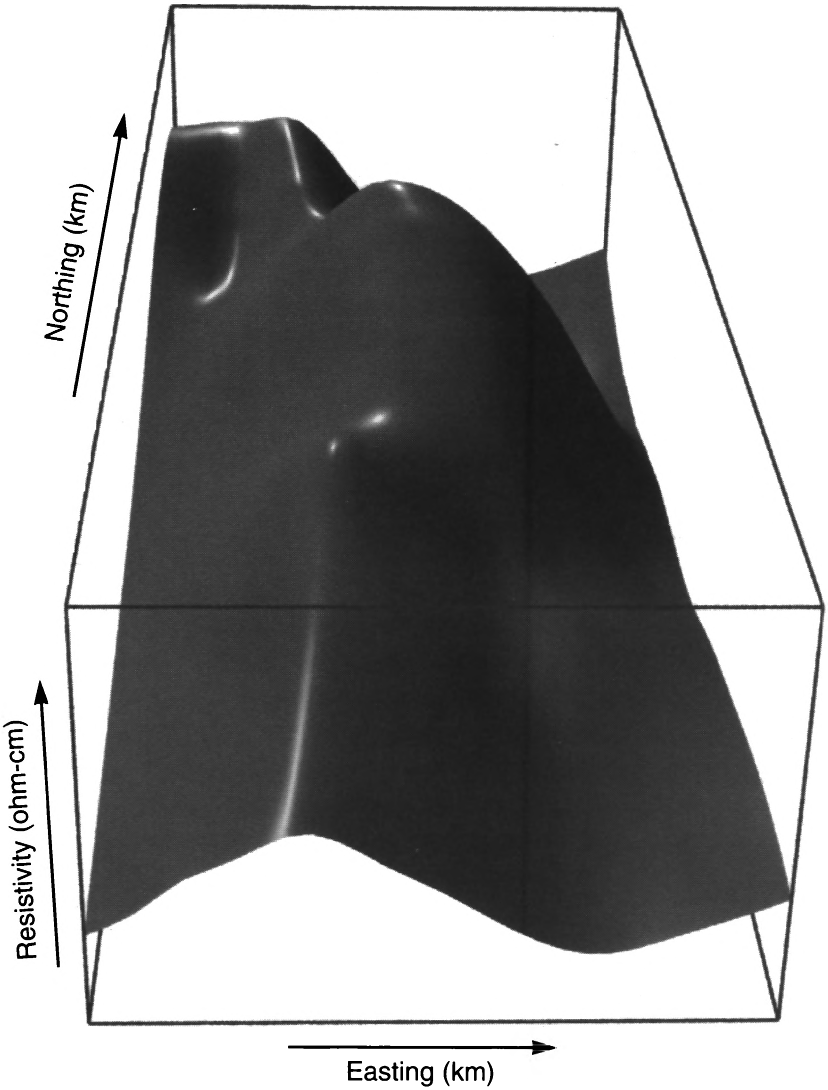
Our visual system is a marvelous device. It provides a perception of depth by processing visual information with extraordinary algorithms [61, 68]. There is the information in the disparity of the two views provided by the two eyes. There is pictorial information, the information in a photograph that allows depth perception; this includes perspective foreshortening and occlusion. Another pictorial cue is the shading of a surface that results from the light sources that illuminate it. And there is the information in motion, both the motion of viewed objects and the motion of the head.

The wireframe renderings presented in Section 4.10 provide depth perception through perspective foreshortening and occlusion. For relatively simple surfaces — for example, the galaxy surface that was displayed in Section 4.10 — wireframe rendering provides an adequate perception of depth. But for more complicated surfaces, a graphics environment that provides both direct manipulation and shading can increase our perception of effects. Figure 4.69 shows a rendering of the resistivity surface with foreshortening, occlusion, and shading; the view was chosen by initial exploration of the surface in a direct manipulation environment.

Shading can provide a substantial increase in our perception of depth. But shading requires many decisions [42, 79]. There is the reflective properties of the surface. It can have a dull matte finish or a shiny finish. The latter results in specular reflection, or shiny areas on the surface. There is the color of the surface. There is the number, color, intensity, and location of point light sources. There is the color and intensity of ambient light. Figure 4.69 has ambient light and three point sources. The color of the lighting is white, and the surface is gray and shiny. The lengths of the northing and easting box lengths have been chosen to make the number of data units per cm the same on both scales.

The chief benefit of direct manipulation is an ability to control the orientation of a surface in a natural way, moving it around as we might a real object held in our hands. This nicely replaces the static method described in Section 4.10 of showing multiple views that form a rotation about the function axis of 360°. Furthermore, motion can strengthen our perception of depth.

Figure 4.69 provides a vivid portrayal of properties of the resistivity surface. The two principal features are the depression and rise to a plateau in the background, and the two peaks in the foreground and midground.



4.69 The fitted resistivity surface is rendered with perspective foreshortening, occlusion, and shading. No tick marks and tick mark labels are drawn since table look-up from this display would be ineffective. (The units are nevertheless shown to remind us of the definitions of the variables.) But the orientation of the surface matches the positioning of the color level plot in Figure 4.68, so as discussed in Section 4.10, the level plot can be used for table look-up. We simply match features found here with those of the level plot, and then use the level plot for the table look-up.

4.14 *Coplots vs. Factor-Plane Methods*

The visualization tools in this chapter can be divided into two groups: conditioning methods and factor-plane methods. The first group — coplots and brushing — employed conditioning on a factor. For the second group — level plots, contour plots, and 3-D plots — the plane of the two factors appeared in the visualization with the factors as coordinate axes.

Conditioning methods are particularly informative when it is natural to study conditional dependence, and when the behavior of the information is characterized in a straightforward way by conditional statements. This was the case for the rubber data, the ethanol data, and the perception data. Factor-plane methods are not nearly as informative in such cases. For example, for the ethanol data, it is informative to understand the effect of compression ratio with equivalence ratio held constant, and to understand the effect of equivalence ratio with compression ratio held constant.

But if conditional behavior is neither particularly interesting nor simple, then factor-plane methods can do better. One test is to ask whether it is just as interesting to see the surface along line segments that are oblique to the coordinate axes of the factors as it is to see the surface along segments parallel to the axes. If the answer is "yes", factor-plane methods are likely to be more useful. This is frequently the case for spatial data. If the answer is "no," coplots are likely to be more useful. The galaxy and soil data are two examples where the answer is "yes", so factor-plane methods were used to visualize the data, although special structure in both data sets was exploited by conditioning methods.

4.15 *Visualization and Probabilistic Inference*

Sometimes, when visualization thoroughly reveals the structure of a set of data, there is a tendency to underrate the power of the method for the application. Little effort is expended in seeing the structure once the right visualization method is used, so we are mislead into thinking nothing exciting has occurred. The rubber data might be such a case.

The intensive visualization showed a linearity in hardness, a nonlinearity in tensile strength, an interaction between hardness and tensile strength, and three aberrant observations in a corner of the factor measurement region. It might be thought that anyone analyzing these data would uncover these properties. This is not the case. In the original treatment, the analysts got it wrong [31]. They operated within a paradigm of numerical methods and probabilistic inference for data analysis, and not intensive visualization. They missed the nonlinearity. They missed the interaction. They missed the outliers. In other words, they missed most of the structure in the data.

The ethanol data were treated no better. In the original analysis by the experimenter, a high-order polynomial was fitted to log concentration by least-squares, and probabilistic inference used standard methods that are based on homogeneous errors with a normal distribution [12]. The visualization in Section 4.4 makes it clear that such probabilistic inferences are clearly not valid; the error distribution is strongly leptokurtic and on a log scale, there is monotone spread. In another study of these data, the mistake of taking logs was not made, but the leptokurtosis of the residuals was missed [23]. In a third study, the analyst used numerical statistical methods to find transformations of the response and factors that supposedly removed the interaction between the factors [69]. But the visualization of the data in this chapter shows that it is absurd to try this. When the equivalence ratio is large, concentration is constant as a function of compression ratio; for other values of equivalence ratio, concentration varies linearly as a function of compression ratio with a positive slope. Monotone transformations of the three variables, the only transformations that make sense for such factor-response data, cannot alter this equality and inequality, and therefore cannot remove the interaction. Finally, an analysis with no visualization of the data made the biggest error [17]. It was concluded that NO_X does not depend on C. Our visualization here manifested a dependence.